

Shubham Gupta

Software Developer

CONTACT

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ShubhamGupta
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TECHNICAL SKILLS

Programming

Python · Shell Scripting · C++
JavaScript · SQL · Java

DevOps & Cloud

Docker · Kubernetes · CI/CD
Apache Airflow · Linux
IBM Cloud · Git · GitHub

Backend & APIs

Flask · FastAPI · Node.js
REST APIs · Postman

AI & LLM Engineering

Agentic AI · Ollama · Granite
IBM Bob · OpenAI Codex
ChatGPT · LLMs · AI Agents
Prompt Engineering
Generative AI
AI-Assisted Development

Data Engineering

SQL · YAML · Apache Airflow
Data Validation · Data Pipelines

Frontend

ReactJS · Carbon Design System

Tools & Platforms

Qiskit · DBeaver · CyberDuck

EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science
and Engineering
2019 - 2023

CERTIFICATIONS

IBM DevOps Fundamentals
IBM Agile Training
Industrial IoT Markets and Security

PROFESSIONAL SUMMARY

Software Developer with 3+ years of experience at IBM specializing in **Generative AI, Agentic AI, LLM-powered applications, Python, AI automation, and cloud engineering**. Hands-on experience building **LLM-powered automation systems** using Python, Ollama, Granite, OpenAI models, ChatGPT, and AI agents for intelligent troubleshooting, automated remediation, code analysis, and workflow automation. Strong experience in **prompt engineering, AI-assisted development, REST APIs, Kubernetes, Docker, CI/CD, Apache Airflow, and cloud-based data processing**. Proven track record of improving production pipeline efficiency by **50%** and contributing to a major data-platform modernization initiative delivering approximately **\$3B in cost savings**.

EXPERIENCE

IBM

Software Developer

Aug 2023 - Present

Bengaluru, India

- Built and deployed a **Jira analytics dashboard** using **Python, Flask, ReactJS, Carbon Design System, and Jira REST APIs**, integrating Jira data for analytics and visualization through a **Docker-based CI/CD pipeline**.
- Engineered an **LLM-powered AI automation system** for Kubernetes operations using **Python, Ollama, and Granite**, integrating AI-driven troubleshooting and automated remediation workflows for process retries, basic issue resolution, and job-failure handling, reducing manual operational intervention.
- Designed and implemented a **multi-source, multi-table data validation framework** in Python, enabling automated data-quality checks across heterogeneous datasets; developed a configuration-based **trend analysis solution using NumPy and Pandas** to generate actionable insights from validated data.
- Optimized and maintained **Apache Airflow DAG-based data pipelines**, performing pipeline debugging, performance optimization, and failure resolution to improve execution efficiency by **50%** and reduce end-to-end runtime from **15 hours to 7.5 hours**.
- Built and maintained **Apache Airflow data-validation workflows** using the **Unified Ingestion Framework (UIF)**, implementing **SQL-driven YAML transformations** to automate data processing, validation, and reporting.
- Contributed to the **sunsetting of BPDM**, a legacy intermediate data pipeline/table that had operated for approximately **6 years**. Rerouted data flows to directly ingest data from legacy sources into the cloud, eliminating an unnecessary intermediate layer, reducing data duplication and invalid records, improving downstream data quality, and contributing to an estimated **\$3B in cost savings**.
- Leveraged **AI-assisted development tools including IBM Bob, OpenAI Codex, ChatGPT, Ollama, and Granite** for code analysis, automated code review, debugging, optimization, and development across **Python, JavaScript, and Shell scripting**, improving engineering productivity and code quality.

IBM

Software Developer Intern

Jan 2023 - Jul 2023

Bengaluru, India

- Developed a **full-stack architecture diagram generator** using **ReactJS and Flask** that automatically generated system architecture diagrams from Excel-based inputs and deployed the application on **IBM Cloud**.

PROJECTS

Quantum Circuit Simulator

Apr 2025 - Sep 2025

 [GitHub](#)

- Built a web-based quantum circuit simulator using Qiskit AER, FastAPI, Python, and ReactJS.
- Containerised the application using Docker for streamlined deployment and scalability.

RasqBerry Pi

Jan 2025 - Aug 2025

 [GitHub](#)

- Integrated quantum simulations and implemented installation checks using Python and Shell scripting to prevent redundancy and improve stability.